

IMPLEMENTATION OF RSA

AIM:

To write a C program to implement the RSA encryption algorithm.

ALGORITHM:

STEP-1: Select two co-prime numbers as p and q.

STEP-2: Compute n as the product of p and q.

STEP-3: Compute $(p-1)*(q-1)$ and store it in z.

STEP-4: Select a random prime number e that is less than that of z.

STEP-5: Compute the private key, d as $e * \text{mod-1}(z)$.

STEP-6: The cipher text is computed as $\text{message} * \text{mod } n$. STEP-7: Decryption is done as $\text{cipherdmod } n$.

PROGRAM:

```
#include <stdio.h>
```

```
long long modExp(long long base, long long exp, long long mod)
```

```
{    long long result = 1;    base
```

```
= base % mod;
```

```
    while (exp > 0)
```

```
    {        if (exp % 2 == 1)            result =
```

```
(result * base) % mod;
```

```
        exp = exp / 2;        base = (base *
```

```
base) % mod;
```

```
    }
```

```
    return result;
```

```
}
```

```
int gcd(int a, int b) {    while  
(b != 0)  
    {        int t = b;        b  
= a % b;        a = t;    }  
return a; }
```

```
int modInverse(int e, int phi) {    int d;  
  
    for (d = 1; d < phi; d++)  
    {        if ((e * d) % phi == 1)  
return d;  
    }
```

```
    return -1; }
```

```
int isPrime(int n)  
{  
    if (n < 2)        return 0;  
  
    for (int i = 2; i * i <= n; i++)  
    {        if (n % i == 0)  
return 0;  
    }  
  
    return 1;  
}
```

```

int main() {    int p,

q;

printf("Enter first prime number: ");    scanf("%d", &p);

printf("Enter second prime number: ");    scanf("%d", &q);

if (!isPrime(p) || !isPrime(q) || p == q)
{
    printf("Invalid prime numbers.\n");    return 0;
}

int n = p * q;
int phi = (p - 1) * (q - 1);
int e;

for (e = 2; e < phi; e++)
{
    if (gcd(e, phi) == 1)
break;
}

int d = modInverse(e, phi);

printf("\nPublic Key = (%d, %d)\n", e, n);    printf("Private Key =
(%d, %d)\n", d, n);

char msg[100];

```

```

printf("\nEnter message (lowercase): ");    scanf("%s", msg);

long long cipher[100];

printf("\nEncrypted Message:\n");

int i;

for (i = 0; msg[i] != '\0'; i++)
{
    int m = msg[i];
cipher[i] = modExp(m, e, n);
printf("%lld ", cipher[i]);
}

printf("\n\nDecrypted Message:\n");

for (i = 0; msg[i] != '\0'; i++)
{
    char ch = (char)modExp(cipher[i], d, n);    printf("%c", ch);
}

printf("\n");

return 0;
}

```

OUTPUT:

```
Output Clear
Enter first prime number: 13
Enter second prime number: 11

Public Key = (7, 143)
Private Key = (103, 143)

Enter message (lowercase): running

Encrypted Message:
49 39 33 33 118 33 38

Decrypted Message:
running

=== Code Execution Successful ===
```

RESULT:

Thus the C program to implement RSA encryption technique had been implemented successfully